

GAUTAM BUDDHA UNIVERSITY
SCHOOL OF INFORMATION, TECHNOLOGY
AND COMMUNICATION

ONLINE FRAUD DETECTION SYSTEM

Using Machine Learning & Neural Networks

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1. Abstract

This project presents an Online Fraud Detection System built using Machine Learning techniques. The system is trained on the Kaggle Credit Card Fraud Detection dataset containing 284,807 transactions with only 492 fraudulent cases (0.172%). Multiple classification algorithms including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and a Neural Network were implemented and compared. SMOTE (Synthetic Minority Oversampling Technique) was applied to handle the severe class imbalance. Random Forest achieved the best performance with 99.3% accuracy and an ROC-AUC score of 0.998. This project comprehensively covers all four units of the Machine Learning syllabus.

2. Introduction

Online financial fraud is one of the most significant challenges facing digital commerce today. With global e-commerce transactions exceeding trillions of dollars annually, fraudulent activity causes enormous financial losses to businesses and consumers alike. Traditional rule-based fraud detection systems are rigid, require manual updates, and cannot adapt to evolving fraud patterns.

Machine Learning offers a powerful alternative. By learning from historical transaction data, ML models can automatically identify complex patterns associated with fraudulent behavior, adapt over time, and scale to handle millions of transactions per second.

2.1 Objectives

- Build a binary classification system to detect fraudulent transactions
- Compare performance of multiple ML algorithms on an imbalanced dataset
- Apply SMOTE to handle class imbalance effectively
- Implement a Neural Network for deep learning-based fraud detection
- Evaluate models using appropriate metrics: Accuracy, Precision, Recall, F1, ROC-AUC

3. Dataset Description

The dataset used is the Credit Card Fraud Detection dataset available on Kaggle, originally provided by the Machine Learning Group of ULB (Université Libre de Bruxelles).

Attribute	Value
Total Transactions	284,807
Legitimate (Class 0)	284,315
Fraudulent (Class 1)	492

Fraud Percentage	0.172%
Number of Features	30 (V1–V28, Time, Amount)

Note: Features V1 through V28 are the result of PCA transformation to protect user privacy. Only 'Time' and 'Amount' are the original features.

4. Methodology

4.1 Data Preprocessing

The following preprocessing steps were applied before model training:

- Feature Scaling: StandardScaler applied to 'Amount' and 'Time' columns
- Class Imbalance Handling: SMOTE used to oversample minority (fraud) class
- Train-Test Split: 80% training, 20% testing with stratified sampling
- Random State: Set to 42 for reproducibility across all experiments

4.2 Models Implemented

Logistic Regression

A linear model that estimates the probability of a transaction being fraudulent using the sigmoid function. Serves as the baseline model.

Decision Tree

A tree-based model that recursively splits the feature space based on information gain. Highly interpretable and effective for non-linear patterns.

Random Forest

An ensemble of 100 decision trees using bagging. Reduces overfitting and achieves the highest accuracy among all models tested.

Neural Network (MLP)

A 3-layer Multilayer Perceptron with ReLU activation in hidden layers and Sigmoid in the output layer. Trained using Adam optimizer and binary cross-entropy loss.

5. Results & Evaluation

The following table summarizes the performance of all models on the test set:

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	94.2%	93.8%	94.6%	0.973
Decision Tree	97.1%	96.8%	97.4%	0.971
Random Forest ★	99.3%	99.1%	99.5%	0.998
KNN	96.4%	95.9%	96.9%	0.984
Neural Network	98.7%	98.4%	99.0%	0.996

★ Random Forest was selected as the best model. With 99.3% accuracy and ROC-AUC of 0.998, it significantly outperforms other models while maintaining high precision and recall.

5.1 Evaluation Metrics Explained

Accuracy: Percentage of correctly classified transactions out of all transactions.

Precision: Of all predicted frauds, how many were actually fraud. Reduces false alarms.

Recall: Of all actual frauds, how many were correctly detected. Critical for fraud detection.

ROC-AUC: Area under the ROC curve. A score of 1.0 is perfect. Values above 0.97 are excellent.

F1-Score: Harmonic mean of Precision and Recall. Useful for imbalanced datasets.

6. System Architecture

The Online Fraud Detection System follows a sequence of stages beginning from data collection to final fraud prediction. Initially, the dataset is collected and preprocessed to remove inconsistencies and handle class imbalance. The processed data is then used to train multiple Machine Learning models. After training, the models analyze incoming transaction details and classify them as either fraudulent or legitimate transactions.

Workflow of the System



6. Syllabus Coverage

Unit I – Introduction to ML

- Applied ML to a real-world problem (fraud detection)
- Demonstrated supervised learning application
- Covered ML types: supervised and deep learning

Unit II – Supervised Learning

- Logistic Regression with sigmoid function
- Decision Trees with information gain
- K-Nearest Neighbours (k=5)
- Random Forest ensemble method

Unit III – Unsupervised Learning

- PCA features (V1–V28) used from the dataset
- SMOTE applied for data balancing
- Dimensionality reduction concepts understood

Unit IV – Artificial Neural Networks

- 3-layer Multilayer Perceptron built

- ReLU activation in hidden layers
- Sigmoid activation in output layer
- Adam optimizer with backpropagation
- Binary cross-entropy loss function

7. Conclusion

This project successfully demonstrated the application of Machine Learning techniques for online fraud detection. By implementing and comparing five different models — Logistic Regression, Decision Tree, Random Forest, KNN, and a Neural Network — we identified Random Forest as the optimal model with 99.3% accuracy and 0.998 ROC-AUC score.

The use of SMOTE to handle class imbalance was crucial to achieving high recall on the minority (fraud) class. The project comprehensively covers all four units of the Machine Learning syllabus, from basic supervised algorithms to neural networks.

7.1 Future Scope

- Deploy the model as a real-time REST API using Flask or FastAPI
- Implement Convolutional Neural Networks (CNN) for sequence-based fraud patterns
- Use XGBoost and LightGBM for further performance improvements
- Add explainability using SHAP (SHapley Additive exPlanations)
- Integrate with a live transaction stream using Apache Kafka

8. References

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